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ORIGINAL PAPER

Therapeutic effects of tDCS on behavioral and cognitive functions in adults diagnosed with attention-deficit/hyperactivity disorder: a systematic review and meta-analysis on randomized controlled trials

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Abstract

The current study aimed to investigate the effectiveness of transcranial direct current stimulation (tDCS) in improving the clinical symptoms and cognitive functions in adults diagnosed with attention deficit/hyperactivity disorder (ADHD) as well as its treatment acceptability. Randomized placebo-controlled trials (RCTs) identified from major databases using the main keywords "tDCS" and "ADHD" were reviewed for improvement in clinical symptoms and cognitive functions in adults diagnosed with ADHD. The outcomes of interest were quantified using effect size expressed as standardized mean difference (SMD) and odds ratios (ORs) for continuous and categorical variables, respectively, with 95% confidence interval (CI). Meta-analysis of seven RCTs including 263 participants (mean age = 31.37) revealed significantly a greater improvement in the symptoms of inattention (SMD=-1.24, 95% CI:-0.84 to -1.63, $p < 0.01$) and cognitive tasks involving attentional performance (SMD= -0.54, 95% CI: -1.07 to -0.01, $p = 0.04$) in the tDCS group compared with the control group, but no significant difference in the symptoms of hyperactivity/impulsivity ($p = 0.19$) and inhibitory control ($p = 0.17$) between the two groups. There was also no significant difference in dropout rate between adults receiving tDCS and their controls ($p = 0.07$). Our study supported the use of tDCS for improving the self-reported symptoms of inattention and objective attentional performance in adults diagnosed with ADHD. However, the limited number of available trials hindered a robust investigation into the parameters required for establishing a standard protocol such as the optimal location of electrode placement and treatment frequency in this setting. Further large-scale studies are warranted to address these issues.

Keywords tDCS · Attention-deficit/hyperactivity disorder · ADHD · Adults and meta-analysis

Introduction

Although attention deficit/hyperactivity disorder (ADHD), which is characterized by typical behavioral symptoms of inattention and/or hyperactivity/impulsivity [41], is one of the most well-known neurodevelopmental disorders in childhood [43], its significance in adults is less well addressed. In fact, up to 60% of individuals diagnosed with this condition exhibit persistent symptoms till adulthood and as high as 41% fit both symptom and impairment diagnostic criteria for adult ADHD [37]. The prevalence of adult ADHD has been reported to be about 3% [38]. Importantly, a high proportion of adults with ADHD has also been found to be afflicted with other concomitant psychiatric disorders [24]. Adults with ADHD often experience additional health complications [15] as well as associated mortality risk [12]

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compared with the general population. For instance, a significantly elevated risk of accidental injury leading to emergency department visits [9] may contribute to an increase in mortality in this population [27]. Therefore, ADHD treatment in adulthood not only aims at improving the quality of life [1] but also targets the associated health and accidental risks [27].

Unlike treatment of ADHD in children for whom evidence in favor of pharmacotherapy is better established, evidence supporting pharmacological treatment for ADHD is less clear in the adult population [10]. Therefore, despite the satisfactory tolerance and general belief in the safety of pharmacological interventions for ADHD, the associated health risks and side effects are still important concerns in adults [5, 17]. Moreover, a considerable portion of adult patients did not respond well to psychostimulant medications [34]. On the other hand, the multi-faceted therapeutic strategy for adult ADHD encompasses counseling, psychoeducation, supportive problem-directed treatment, coaching, behavioral intervention, and family therapy [40]. However, except cognitive behavioral therapy, evidence remains insufficient to support the effectiveness of other types of psychosocial interventions in adult ADHD treatment [39].

Transcranial direct current stimulation (tDCS), which involves the application of a low-intensity current over the scalp, has been reported to regulate activity in cognitive control circuits [33], through the modulation of cortical excitability and induction of neuroplasticity [13]. Notwithstanding some promising results reported in prior studies investigating the therapeutic effects of tDCS against ADHD in adults, their findings were inconsistent [2–4, 8, 11, 16, 26]. Likewise, although a few previous meta-analyses suggested the effectiveness of tDCS against the symptoms of ADHD, none of them focused on the adult population [7, 42]. It is noteworthy that the rating tools for the clinical symptoms of ADHD are fundamentally different between adults and children; while the assessment of adult symptoms was mainly based on self-rating scales, the evaluation of symptoms in children is primarily based on external observers [32]. In addition, the clinical presentations of ADHD in adults, which can be quite different from those of their pediatric counterparts, cannot be simply viewed as an extension of childhood symptoms [25]. Despite the therapeutic significance of ADHD in adults, relevant studies remain scarce. The robustness of evidence supporting the beneficial effect of tDCS against inattention from subgroup analysis of the adult population in a previous meta-analysis was tarnished by the limited number of included trials as well as a high heterogeneity (I^2 = 80%) probably attributable to a mix of self-reported and performance-based data for analysis [42]. Taking into account that self-rating scales may be more susceptible to placebo effects than performance-based assessment as reported in a previous drug efficacy study among psychiatric patients [22], we examined the effects of tDCS treatments by separating self-reported data and those acquired from performance-based tasks. Moreover, none of the previous meta-analyses investigated the treatment acceptability of tDCS in the adult population with ADHD [7, 42].

Therefore, the present meta-analysis, which targeted randomized controlled clinical trials (RCTs), aimed at renewing current evidence regarding the effectiveness of tDCS in improving the clinical symptoms and cognitive functions in adults diagnosed with ADHD as well as its treatment acceptability.

Methods

Study design and registration

The current meta-analysis was presented in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [30]. The protocol of the study was registered in the international prospective register of systematic reviews (PROSPERO CRD 42025644112).

Search strategy and eligibility criteria

Randomized controlled trials (RCTs) focusing on the effectiveness of tDCS in improving behavioral and cognitive functions of adults diagnosed with ADHD were acquired through screening electronic databases, including PubMed, Embase, Cochrane CENTRAL, ScienceDirect, and ClinicalTrials.gov from inception to Feb 8, 2025 using the main keywords of "ADHD + tDCS" based on appropriate search strategies without imposing restrictions on the language and geographic origins of publication (eTable1). The reference lists of the retrieved literature were further scrutinized for potentially eligible articles to be included. Each article was screened for eligibility based on its recruited population, intervention, comparator, and outcomes (PICO): (1) Population: adults regardless of gender diagnosed with ADHD enrolled in a RCT, (2) Intervention: tDCS, (3) Comparator: non-tDCS interventions (e.g., sham-tDCS or waitlist), and (4) Outcome: alterations in behavioral symptoms or cognitive functions. We excluded studies that (1) were not RCTs; (2) did not use tDCS as an intervention, (3) did not recruit adults diagnosed with ADHD, or (4) provided no information about behavioral or cognitive functions as their study outcomes.

Data extraction and quality assessment

Following independent screening of the titles and abstracts of the identified articles according to the preset keywords and strategies (eTable1), the same two authors (YH Wen and CK Sun) continued to extract data on study characteristics and outcomes. Disagreements regarding study eligibility or data to be included were resolved through discussion with a third author (WF Pan). Kappa coefficients were adopted to assess inter-rater reliability [30]. The corresponding authors of articles that had missing data for analysis were contacted through electronic mails in an attempt to retrieve the necessary information. While the quality of the included studies was appraised with the Cochrane's "risk of bias" assessment tool [21], the level of evidence of each outcome was assessed with the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) framework [19]. Disagreements between the two authors on the risk of bias or level of evidence were resolved through discussion.

Data synthesis and analysis

The primary outcome of the current study was changes in ADHD-associated behavioral symptoms evaluated with standard behavioral rating scales including ADHD Self-Report Scale (ASRS), adult ADHD Self-Report Scale (CASRS), or cognitive functions related to ADHD such as those focusing on attentional function (i.e. continuous performance test) and inhibitory control (i.e. Stroop test). The secondary outcome was tDCS treatment acceptability assessed by the numbers of withdrawn participants in the tDCS and control groups. The outcomes of interest were quantified using effect size (ES) expressed as standardized mean difference (SMD) and odds ratios (ORs) for continuous and categorical variables, respectively, with 95% confidence interval (CI). Review Manager 5 (RevMan5.4; Copenhagen: The Nordic Cochrane Center, The Cochrane Collaboration, 2014) was utilized for all statistical analyses. While we adopted the generic inverse-variance method for analyzing continuous data, the Mantel-Haenszel method was used for categorical variables. To examine the robustness of study outcomes, sensitivity test with leave-one-out approach was used. The probability of publication bias and heterogeneity were appraised with funnel plot and *I*-squared test, respectively. The impact of target brain region and type of control (e.g. waitlist or sham-tDCS) on the efficacy of tDCS treatment was assessed with subgroup analysis. A probability value (p) less than 0.05 was considered statistically significant for all study outcomes.

Results

Eligible studies and characteristics

The literature selection process was reported in compliance with the PRISMA guidelines [31] (Fig. 1). Of the 698 articles retrieved from the electronic databases using the preset keywords and search strategies (eTable2), 675 were deemed ineligible following title and abstract screening. A full-text scrutiny of the remaining 23 RCTs yielded seven eligible studies including 263 participants for the current meta-analysis [2–4, 8, 11, 16, 26] (Fig. 1). The age of the participants ranged from 18 to 67 years with a mean age of 31.37 (Table 1). In respect of study design, five studies adopted a parallel design [3, 4, 8, 11, 26], while two were cross-over trials [2, 16]. The median number of treatment sessions of tDCS treatment was five and the median duration of treatment was seven days. Regarding tDCS location, three studies positioned the anode at F4 [3, 8, 26], three at F3 [2, 4, 11], one targeted both sides of the prefrontal cortex (PFC) with different treatment arms [16], while all placed their cathodes at corresponding positions of the anodes on the contralateral side of the brain [2–4, 8, 16, 26]. The countries of origin of the eligible RCTs included USA [2, 26], Brazil [8, 11, 26], Turkey [3], Iran [4].

Risk of bias appraisal

The Cochrane Collaboration's tool demonstrated a low risk of bias in randomization sequence in most trials except for some studies that were given an unknown risk due to a lack of information about allocation concealment (Fig. 2). Likewise, most studies were deemed to have a low risk of both detection and performance bias because of their sham-control designs with the exception of one that recruited controls without receiving sham-tDCS [4]. Nevertheless, the correct recognition of sham-tDCS in a high proportion of adult participants as previously reported has underscored the questionable blinding effects among sham-controls [18]. Reporting bias and attrition bias were considered low in all studies because they all targeted symptoms or cognitive functions of ADHD with a low drop-out rate.

Primary outcome

The current meta-analysis showed significant improvement in the clinical symptoms of inattention (SMD=-1.24, 95% CI:-0.84–-1.63, $p < 0.01$, three studies, 121 participants) (Fig. 3A) and cognitive tasks involving attentional performance (SMD= -0.54, 95% CI: -1.07–-0.01, $p = 0.04$, three

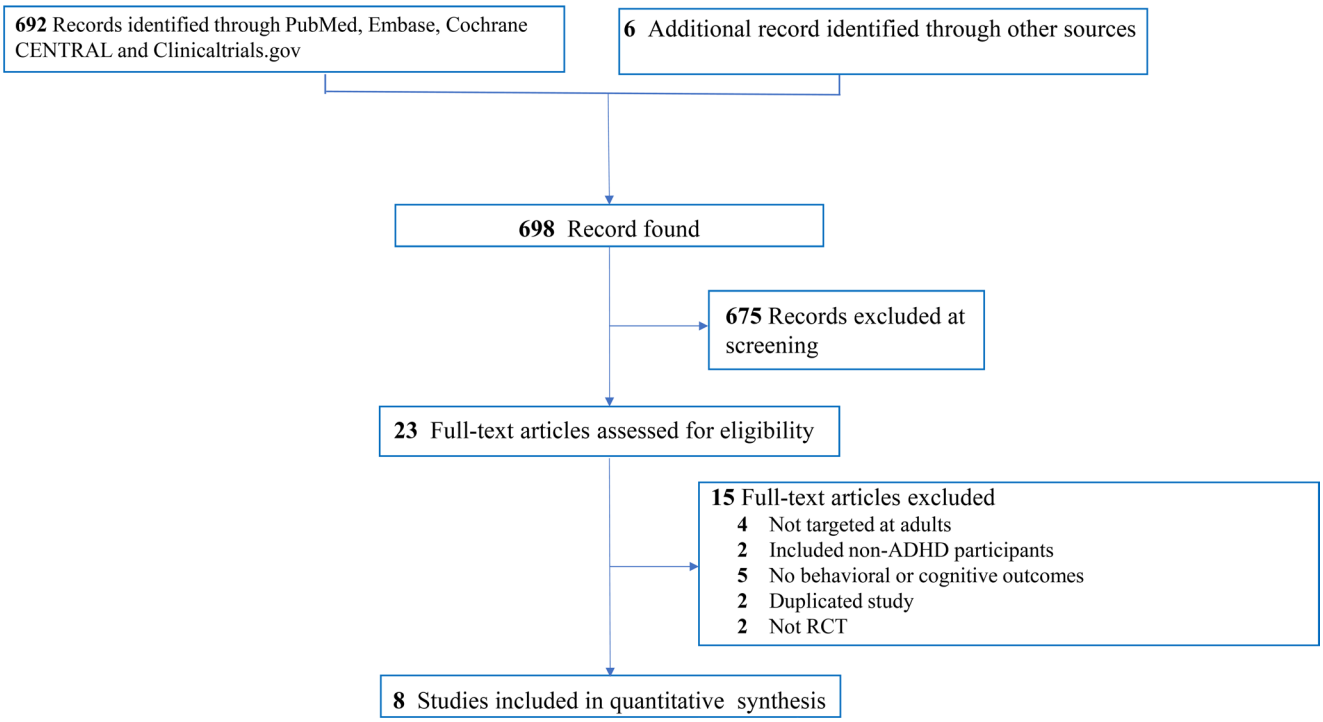


Fig. 1 PRISMA diagram of identifying eligible studies. *ADHD* Attention deficit hyperactivity disorder, *RCT* randomized controlled trials

studies, 136 participants) (Fig. 4A) in adults diagnosed with ADHD receiving tDCS compared to their controls. No significant heterogeneity (I^2 = 0% and p = 0.71; I^2 = 51% and p = 0.13) and publication bias on funnel plot inspection (eFigs. 1 and 2) were noted across the included studies. Sensitivity analysis also demonstrated no significant impact of any study on the outcomes regarding the symptoms of inattention, but loss of statistical significance after removal of any one of two studies [3, 4] on the outcomes concerning attentional performance. Due to the limited availability of suitable trials, subgroup analyses could be conducted only on studies targeting the right PFC and those using sham-tDCS as controls focusing on improvement in the symptoms of inattention (SMD=-1.18, 95% CI:-1.66–-0.71, p < 0.01, two studies, 81 participants, and SMD=-1.18, 95% CI:-1.66–-0.71, p < 0.01, two studies, 81 participants, respectively). On the other hand, subgroup comparisons between different target brain locations and types of control groups were not possible due to limited availability of eligible trials. Apart from inattention and attentional performance, our results showed no significant difference between adults with ADHD receiving tDCS and their controls in the symptoms of hyperactivity/impulsivity (SMD=-0.24, 95% CI:-0.06–0.12, p = 0.19, three studies, 121 participants) (Fig. 3B) and inhibitory control (SMD= -0.19, 95% CI:-0.47–0.08, p = 0.17, five studies with 228 participants) (Fig. 4B). The level of heterogeneity (I^2 = 0% and p = 0.69; I^2 = 11% and p = 0.22, respectively) and publication bias on funnel plot inspection (eFigs. 3 and 4) across the included studies were non-significant. Sensitivity analysis demonstrated no significant impact of any study on the symptoms of hyperactivity/impulsivity, but inhibitory control became statistically better in the tDCS group than that in the control group after removal of one study [2].

Secondary outcomes

Our secondary analyses on adults diagnosed with ADHD revealed no significant differences in dropout rate between those receiving tDCS and their controls (OR=3.39, 95% CI:0.89–17.35, p=0.07, four studies with 143 participants) (eFigure 5). The level of heterogeneity (I^2=0% and p=0.89) across the included studies was non-significant.

Certainty of evidence

According to the GRADE guidelines, the evidence of our primary outcomes targeting improvement in the symptoms of inattention and hyperactivity/impulsivity, attention and performance, as well as inhibitory control were downgraded to very low because of the limited number of available studies (i.e., only three to five eligible trials) (eTable 3). The evidence for acceptability was also downgraded to very low by reason of the small number of eligible studies (n=4).

Table 1 Summary of characteristics of studies in the current meta-analysis

Study (year)	Diagnosis (Criteria)	Design	Comparison	*N*	Number of Sessions	Duration	Outcome	Mean age (years)	Female (%)	Country
Cosmo (2015)	ADHD (DSM-IV-TR)	RCT	tDCS: Anode-F3; Cathode-F4	30	1	1 day	Inhibitory control – go/no go tasks	32.25 (18–65)	41.67	Brazil
			Sham-tDCS	30						
Cachoeira (2017)	ADHD (DSM-5)	RCT	tDCS: Anode-F4; Cathode-F3	9	5	5 days	Overall – ASRS Inattention – ASRS H/I - ASRS	32.29 (18–45)	52.94	Brazil
			Sham-tDCS	8						
Allenby (2018)	ADHD (DSM-5)	RCT/ Crossover	tDCS: Anode- F3; Cathode-right-supraorbital area	37	3	7 days	Attention functional – CPT Inhibitory control - SST	31.7 (18–65)	25.71	U.S.
			Sham-tDCS	37						
Barham (2022)	ADHD (DSM-5)	RCT	tDCS: Anode-F4; Cathode-F3	11	5	5 days	Attention functional – CPT Inhibitory control - ST	22 (>18)	68.2	Turkey
			Sham-tDCS	11						
Dubreuil-Vall (2022)	ADHD (DSM-5)	RCT/ Crossover	tDCS: Anode - F3/F4; Cathod e – Fp2/Fp1	23	3	N/A	Inhibitory control –SST	31.2 (18–67)	50	U.S.
			Sham-tDCS	23						
Leffa (2022)	ADHD (DSM-5)	RCT	tDCS: Anode-F4; Cathode-F3	32	28	28 days	Overall – CASRS Inattention – CASRS H/I – CASRS	38.3 (18–60)	46.88	Brazil
			Sham-tDCS	32						
Basiri (2023)	ADHD (DSM-5)	RCT	tDCS: Anode-F3; Cathode-F4	20	10	30 days	Inattention – ASRS H/I – ASRS Attention functional – CPT Inhibitory control - ST	23.5 (18–45)	78.75	Iran
			Control	20						

ADHD Attention-deficit/hyperactivity disorder, *ASRS* ADHD Self-Report Scale,*CASRS* Adult ADHD Self-report Scale,*CPT* continuous performance test, *DSM-5* Diagnostic and statistical manual of mental disorders, fifth edition, *DSM-IV-TR* Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision; *H/I* Hyperactivity/impulsivity, *N* number; *N/A* not available, *RCT* randomized controlled trial, *ST* Stroop test, *SST* stop signal test, *tDCS* Transcranial direct current stimulation

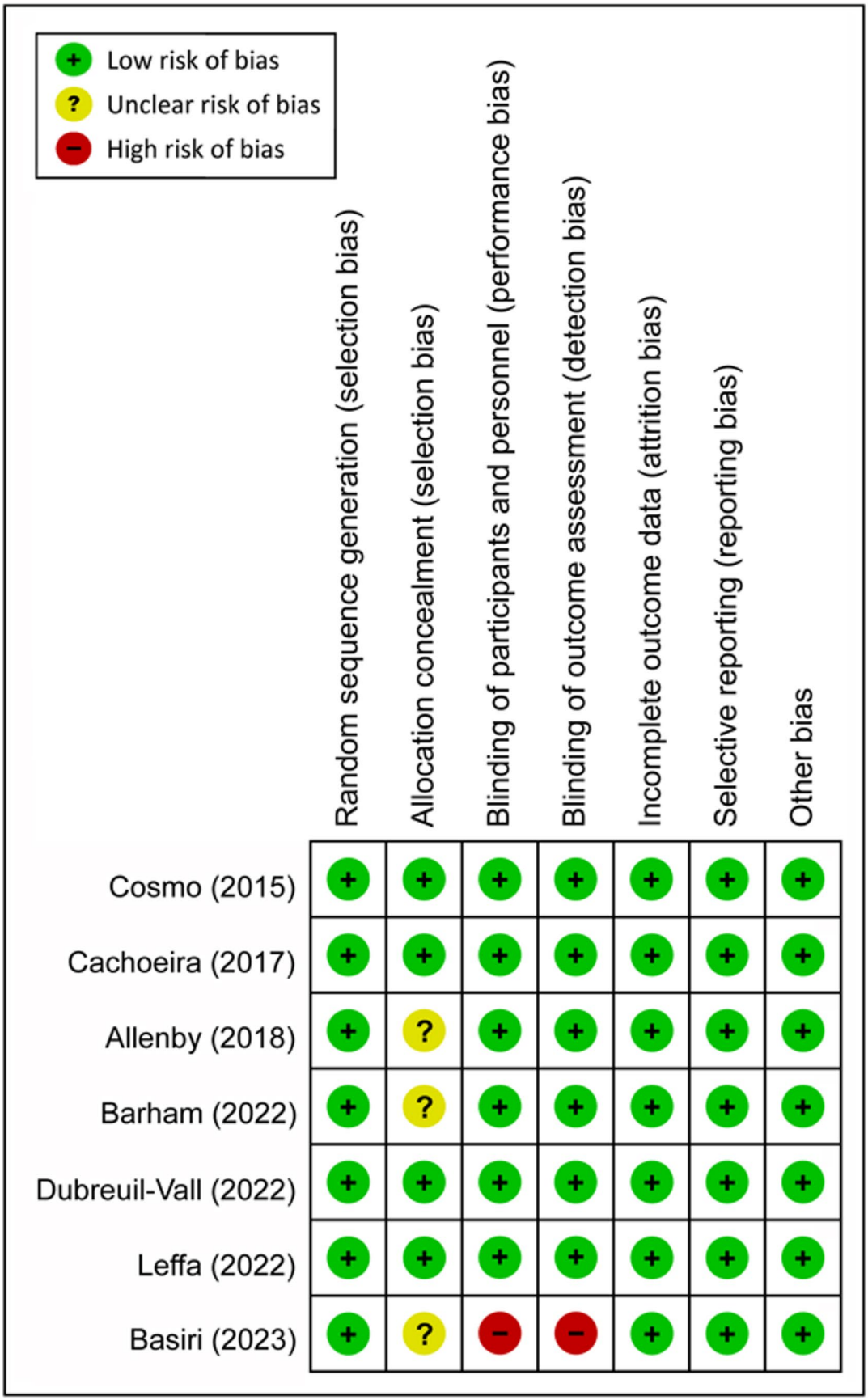

Low risk of bias
Unclear risk of bias
High risk of bias
Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participants and personnel (performance bias)
Blinding of outcome assessment (detection bias)
Incomplete outcome data (attrition bias)
Selective reporting (reporting bias)
Other bias
Cosmo (2015)
Cachoeira (2017)
Allenby (2018)
Barham (2022)
Dubreuil-Vall (2022)
Leffa (2022)
Basiri (2023)

Fig. 2 Risk of bias for eligible studies

Discussion

The present meta-analysis showed effectiveness of tDCS treatment based on subjective perception of inattention symptom alleviation as well as objective improvement in attentional performance in adults diagnosed with ADHD. Our finding was consistent with that of a previous meta-analysis that reported improvement in attentional performance in a subgroup of adult ADHD patients [42]. On the other hand, the level of heterogeneity was much lower in our meta-analysis compared with that in the prior meta-analysis (I^2 = 0% vs. 80%, respectively), suggesting more robust evidence derived from the current study. Our subgroup analysis further indicated the effectiveness of tDCS with the anode placing over the right PFC in improving attention. Another subgroup analysis focusing on studies using sham-controls also supported the therapeutic effects of tDCS. Regarding our secondary outcomes, the present study demonstrated no significant difference in dropout rate between the tDCS and control groups, implying fair acceptability of tDCS treatment. Nevertheless, the certainty of evidence regarding our positive findings on the therapeutic effects of tDCS against the symptoms of inattention and its fair treatment acceptability was rated as very low due to the limited numbers of available trials (i.e., seven trials and 263 adult participants), suggesting uncertain efficacy of tDCS as a standard treatment for ADHD in adult patients that needs to be verified by further large-scale RCTs. In particular, an ongoing multicenter, randomized, double-blind, and sham-controlled study that evaluates the short- and mid-term effects of tDCS may provide further evidence regarding the therapeutic efficacy of tDCS in the adult ADHD population [29].

Notwithstanding the unclear mechanisms underpinning the observed positive impact of tDCS on cognition in neuropsychiatric patients [13], cortical excitability modulation and neuroplasticity induction have been proposed to play a role [33]. Provided that the PFC is pivotal in the executive control of attention [36], it is a frequent target for brain stimulation [6]. Indeed, all studies placed the anode either over the left or the right PFC in our included studies [2–4, 8, 16, 26]. Although our result that showed highly significant effectiveness of tDCS treatment in improving the symptoms of inattention as reflected by a large ES (ES=-1.24) was comparable to that of a previous meta-analysis (ES=-1.03) [42], a notably lower heterogeneity (I^2 = 0%) in the current study than that of the prior meta-analysis (I^2 = 80%) [42], was probably attributable to our separating the data on subjective clinical rating from those on objective performance-based assessment during our analysis. Importantly, our results showed apparently substantial therapeutic benefit in the former (ES= -1.24) but only moderate effectiveness in the latter (ES = 0.54). Indeed, the blinding effect of sham tDCS in adults remained controversial [18]. Therefore, our results and those of others [22] may suggest an increased susceptibility to placebo effects when subjective self-rating scales were used. Hence, despite our findings indicating the effectiveness of tDCS in alleviating the clinical symptoms of inattention and improving cognitive functions among those diagnosed with ADHD, the discrepancy in ES between the two outcomes may suggest the necessity of further studies to include both subjective behavioral rating and objective performance-based parameters to minimize assessment bias. Given the poor blinding effects of sham tDCS in adults [22], our findings may imply higher reliability of cognitive tests for evaluating the therapeutic effects of tDCS on attention than that of subjective symptom ratings. Nevertheless, further large-scale studies are warranted to address this issue.

Due to the limited number of included studies that selected different target regions (e.g., only one study placing the anode over the left PFC investigated clinical improvement after tDCS intervention [26]), we could only conduct subgroup analysis focusing on the placement of anode over the right PFC for improving the symptoms of inattention. Our results demonstrated greater improvement in those receiving tDCS treatment than that in the control group. Despite the known association between ADHD-related inattention and PFC, previous studies suggested more involvement of the right PFC than that of the left [14, 20]. A meta-analysis focusing on the differences in neuroanatomical activation between patients diagnosed with ADHD and healthy controls showed an association between reduced capability for inhibition and decreased activation in the right inferior frontal cortex, as well as a correlation between poor performance in attention tasks and decreased activation in the right dorsolateral prefrontal cortex [20]. Although several regions of the left hemisphere such as the left putamen and the left globus pallidus were also found to be associated with inhibition and inattention, these regions were farther away from the frontal area. Therefore, tDCS treatment of inattention through placing the anode over the right PFC seemed a reasonable approach, which was supported by the findings of our subgroup analysis that demonstrated a link between anode placement over the right PFC and an improvement in the symptoms of inattention. On the other hand, because the therapeutic benefit of tDCS may be attributed to its neuroplastic effect on the functional connectivity of the whole brain [35]; electrical stimulation may exert a generalized influence instead of an effect confined to specific brain regions. Because there was only one study focusing on the left PFC for clinical symptoms of inattention [26], we were unable to compare the therapeutic benefits between studies focusing on the right PFC and those targeting the left PFC.

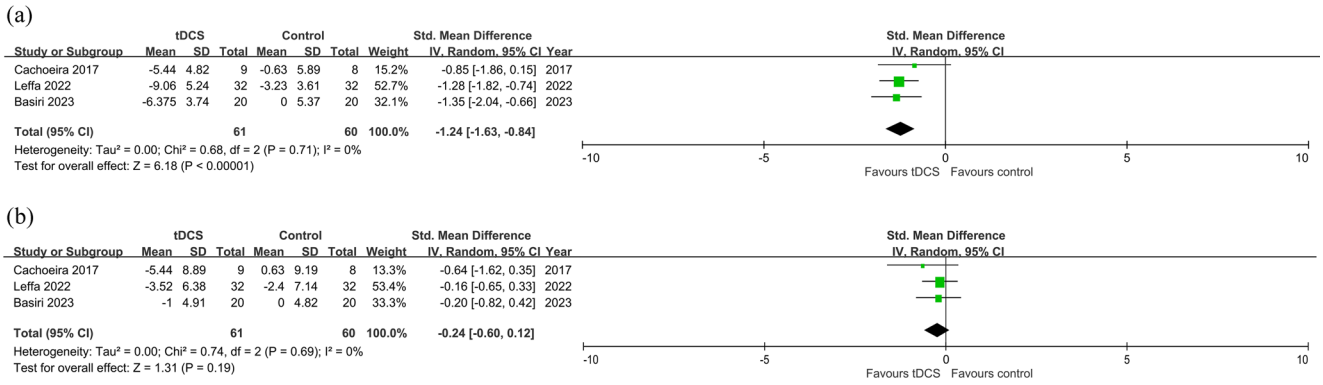


Fig. 3 Forest plot of effect size for comparing the difference in the (a) symptoms of inattention and (b) symptoms of hyperactivity/impulsivity between tDCS and control . *CI* confidence interval, *tDCS* transcranial direct current stimulation, *Std* standardized, *SE* standard error

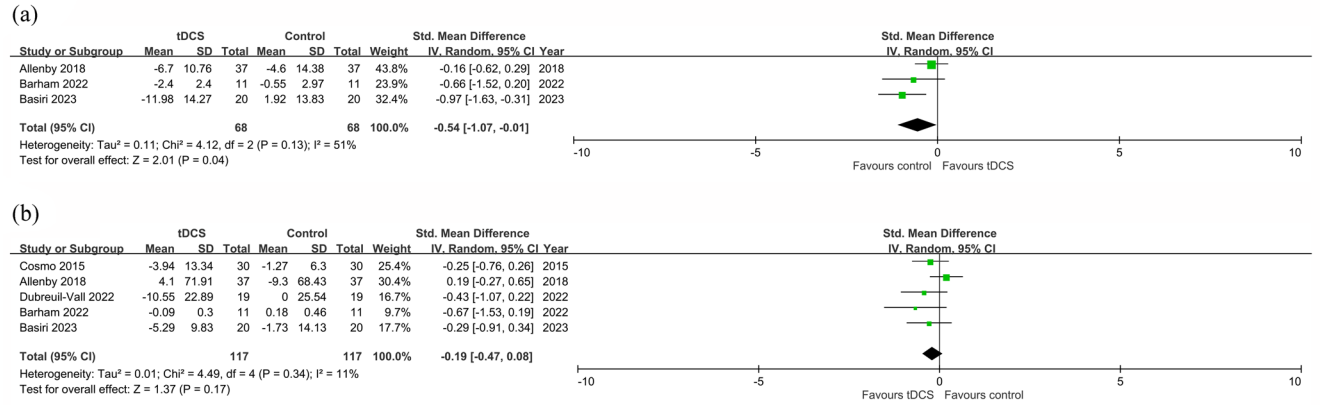


Fig. 4 Forest plot of effect size for comparing the difference in the (a) attentional performance and (b) inhibitory control between tDCS and control. *CI* confidence interval, *tDCS* transcranial direct current stimulation, *Std* standardized, *SE* standard error

Nevertheless, given the evidence showing the predominant involvement of the right PFC in the symptoms of inattention [20], as well as the reported variation in the outcome on performance enhancement in category learning with the location of tDCS anode placement [23], further studies are needed to identify the appropriate locations of anode and cathode placement to maximize its therapeutic benefit.

The current study showed that tDCS was effective in improving neither the behavioral rating of hyperactivity/impulsivity nor the performance-based tasks involving inhibitory control. In contrast to the notable discrepancy in ES between subjective inattention ratings and objective attentional assessments in our findings, such an absence of difference in ES between the subjective hyperactivity/impulsivity ratings and objective performance-based tasks may be attributable to a blunted subjective perception of improvement due to a lack of awareness of the symptoms of hyperactivity/hyperactivity compared to those of inattention in the adult population [28]. Regarding objective assessment, despite the lack of a meaningful impact of tDCS on objective inhibitory control when all studies were considered, sensitivity analysis demonstrated significantly greater improvement in inhibitory control in the tDCS group compared with the controls after exclusion of one study [2], challenging the robustness of this finding. Taking into account the absence of difference in treatment-related parameters, including number of sessions, target location, and protocol of tDCS (e.g., energy, duration of stimulation) between that trial [2] and others [3, 4, 16]], the reason for poorer therapeutic effect in the former remains unclear. Together with limited availability of eligible trials, the reliability of our outcome on inhibitory control was inadequate to exclude potential therapeutic effects of tDCS on inhibitory control in adults diagnosed with ADHD. Likewise, our results on improvement in the symptoms of hyperactivity/impulsivity, which were derived only from three trials, were not robust enough to exclude the treatment potential of tDCS against such symptomatology in adult ADHD patients.

Our secondary outcome investigating treatment acceptability showed no significant difference between the tDCS and control groups, suggesting fair acceptability associated with tDCS treatment. However, the lack of information on dropout due to side effects of tDCS in all of the included trials precluded the possibility of conducting a meta-analysis

regarding the tolerability of tDCS treatment that requires further studies to elucidate.

The present study had several limitations that need to be taken into account. First, our results derived from merely seven trials and a limited number of participants could not provide solid evidence to support the therapeutic effectiveness of tDCS in improving either inattention or the symptoms of hyperactivity and impulsivity in adults with ADHD. Second, our inclusion of trials only from North and South America as well as Turkey may not justify the extrapolation of our findings to populations of other geographical or ethnic backgrounds (e.g., Asians or Europeans). Third, despite our intention of investigating the impact of stimulating different target brain regions on the therapeutic effects of tDCS, we could only conduct subgroup analysis on the right PFC which is the prevalent target region across the studies. As a result, the difference in treatment effectiveness between the right and left PFC as well as that between PFC and the other regions of the brain remains to be elucidated. Finally, variations in the protocol of tDCS treatment (e.g., the number of sessions and the duration of follow-up) coupled with the availability of only five trials that ruled out the possibility of a meta-regression analysis precluded an in-depth investigation into the impact of these factors on therapeutic outcomes. Further large-scale investigations are warranted to establish a standard protocol on the therapeutic use of tDCS in adults diagnosed with ADHD.

Conclusions

Our study supported the use of tDCS for improving the self-reported symptoms of inattention and objective attentional performance in adults diagnosed with ADHD. However, the limited number of available trials hindered a robust investigation into the parameters required for establishing a standard protocol such as the optimal location of electrode placement and treatment frequency in this setting. Further large-scale double-blind sham-controlled clinical trials that include assessments of self-reported symptoms and performance-based tasks both immediately after interventions and during follow-up periods as well as comparisons of the efficacy of tDCS targeting different brain locations are warranted to address these issues.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00406-025-02162-1.

Author contributions YH Wen, WF Pan and CK Sun conceived and designed the study; YH Wen, and CK Sun contributed to data extraction; YS Cheng, and WF Pan analyzed the data; YH Wen, WF Pan, and CK Su and KC Hung were major contributors to writing the manuscript. All authors contributed sufficiently to this work. All authors read and approved the final manuscript.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication All authors agreed the possible publication of our article on European Archives of Psychiatry and Clinical Neuroscience.

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